

Bayesian MCMC Validation of the Structural Self-Energy Expansion Framework (SSEE):

Model Comparison against Λ CDM and CPL
using DESI DR2, Planck 2018, and Galaxy-Cluster Mass Data

Mike Edison Almeida Vallejo

Independent Researcher

mike.almeida1721@gmail.com

ORCID: [0009-0008-2195-7836](https://orcid.org/0009-0008-2195-7836)

April 2026

Abstract

We present a Bayesian validation of the dynamical sector of the Structural Self-Energy Expansion (SSEE) framework [Almeida Vallejo, 2026] against DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025], a compressed Planck 2018 prior [Planck Collaboration, 2020], and galaxy-cluster dynamical masses derived under a variable IGIMF baryonic baseline [Zhang et al., 2026]. Using geometric constants fixed algebraically by π and the golden ratio φ — with H_0 and $\Omega_b h^2$ as the only sampled parameters — we compare the model against Λ CDM and a free CPL parameterisation via posterior constraints, information criteria, and predictive checks, using EMCEE [Foreman-Mackey et al., 2013, Goodman & Weare, 2010] with $N_{\text{walkers}} = 100$, $N_{\text{steps}} = 25,000$ (2.5×10^6 total evaluations), and a same-bin correlated block-diagonal covariance approximation for the DESI DR2 BAO measurements; convergence diagnostics (including $N_{\text{eff}} = 4,402$ effective samples for SSEE) are reported in Table 5. The SSEE prediction $(w_0, w_a) = (-0.840, -0.670)$, fixed algebraically by the closed dictionary [Almeida Vallejo, 2026b] and not fitted to the data, lies within 0.24σ of the official DESI DR2 w_0 - w_a CDM posterior (DESI+CMB+Pantheon+; arXiv:2503.14738, eq. 26) in the w_0 - w_a plane, and within 1.2 – 1.5σ of the DESY5/Union3/no-SN combinations (a parameter-free post-diction). The H_0 posterior is $67.82^{+0.41}_{-0.41} \text{ km s}^{-1} \text{ Mpc}^{-1}$, obtained with the global rate of the screening cascade as the H_0 prior ($H_0^{\text{glob}} = H_{\text{SH0ES}}(1 - f_{\text{screen}}) = 67.962 \pm 0.968$; Papers 3, 9–10, §3.2), 0.33σ from that prediction. The cluster-mass sector achieves $\chi_r^2 = 0.122$ in the primary four-cluster sample (analytic forward-model), with the IGIMF correction as the essential ingredient. With (w_0, w_a) fixed algebraically and one free parameter H_0 , SSEE is favoured by $\Delta\text{BIC} = -6.77$ over Λ CDM and -5.65 over CPL, and predicts the DESI DR2 test set better in leave-one-out cross-validation — so the w_0 - w_a prediction is genuinely preferred by current BAO data. The background geometry uses the *total* gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$ (ω_m -direct; no matter factor, OP-8 dissolved), for which the SSEE sound horizon matches Λ CDM ($r_d^{\text{SSEE}}/r_d^{\Lambda\text{CDM}} = 1.002$) and Ω_m agrees

with Planck at 0.88σ . The saturation $s_m = 1 + w_0 = 0.160050$ is a number of the dark-energy equation of state, not a density, and does not enter the background geometry.

Note on the background matter density: Throughout, the background geometry ($E(z)$, the sound horizon r_d , and all BAO distances) uses the *total* gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$, built piece-by-piece from SSEE algebra (ω_m -direct; no matter factor, OP-8 dissolved; Paper 3). The saturation $s_m = 1 + w_0 = 0.160050$ is a number of the dark-energy equation of state, *not* a matter density, and never enters $E(z)$. With the total density the SSEE sound horizon matches Λ CDM and model selection yields a single, unambiguous figure: $\Delta\text{BIC} = -6.77$ favouring SSEE over Λ CDM and -5.65 over CPL (Section 8).

Contents

1	Introduction	4
2	SSEE Model Summary	5
2.1	Algebraic constants	5
2.2	Friedmann background (ω_m -direct)	5
2.3	Cluster mass formula	6
2.4	The sound horizon under the ω_m -direct reframe	6
3	Datasets and Likelihoods	7
3.1	DESI DR2 BAO	7
3.2	SSEE-self-consistent H_0 prior: the ω_m -direct CMB anchor	8
3.3	Galaxy-cluster masses	9
4	Bayesian Framework and Priors	10
5	MCMC Implementation	10
6	Results: Cosmological Sector	11
6.1	Posterior parameters	11
6.2	Position in the w_0 - w_a plane	13
6.3	Sound horizon and H_0 posterior	15
6.4	Posterior predictive check: $H(z)$	17
6.5	BAO residuals	19
7	Results: Galaxy-Cluster Sector	19
8	Model Comparison	21
9	Robustness Checks	21
10	Discussion	23
10.1	The dynamical sector and the background	23
10.2	Growth of structure	24
10.3	Falsifiability	24
11	Conclusions	24
A	Rigorous Bayesian Model Comparison: DIC and BIC	25
B	Nested Bayes Factor and Predictive Cross-Validation	26

1 Introduction

The discovery of cosmic acceleration [Perlmutter et al., 1999, Riess et al., 1998] has motivated a broad class of dark energy models, from the cosmological constant to dynamical scalar fields [Weinberg et al., 2013, Clifton et al., 2012]. Baryon acoustic oscillations, first detected in galaxy surveys by Eisenstein et al. [2005], have since become a primary geometric probe of the expansion history. The SSEE framework [Almeida Vallejo, 2026] is a **minimal-parameter phenomenological framework** that encodes the cosmological background as algebraic combinations of π and $\varphi = (1 + \sqrt{5})/2$. The background sector $(w_0, w_a, \Omega_{\text{DE}}, \Omega_m)$ tested in this paper carries zero dimensionless fitted parameters, and no WIMP, QCD-axion, or SUSY dark-matter particle is introduced; phenomenological extensions in the dark-matter sector (Paper 6) introduce ansätze tracked as open problems (see Paper 1, §1.3, and `OPEN_PROBLEMS.md`). Paper 1, Appendix A provides a complete Lagrangian (k-essence + Israel-Stewart viscosity) derived algebraically from $\{\varphi, \pi\}$, with zero free parameters at the EFT background level. Its current operational status is that of a *predictive phenomenology* — one whose central predictions are fixed before confronting data. Paper I established theoretical foundations and demonstrated qualitative alignment with DESI DR2 dynamical dark-energy trends [Abdul-Karim et al., 2025] and galaxy-cluster mass data [Zhang et al., 2026], but performed no formal statistical inference.

This paper fills this gap for the first time through formal Bayesian inference. Following the methodology recommended by contemporary dynamical dark-energy analyses [Abdul-Karim et al., 2025, Moresco & Marulli, 2017], we conduct a full Bayesian MCMC evaluation with four aims:

1. Quantify whether SSEE remains competitive under formal Bayesian inference against Λ CDM and free CPL alternatives.
2. Identify which physical ingredients reproduce cluster masses.
3. Document why the naive $\Omega_{\text{DE}}^{\text{SSEE}} \approx 0.840$ vs $\Omega_{\Lambda} \approx 0.685$ comparison is a category error (s_{DE} vs. density fraction) and localise its origin in the retired two-sector framing (see §2.4).
4. Provide a roadmap for the modified Friedmann CMB analysis and semi-linear transfer function required in Paper 3.

The framing is explicit: *the goal of this paper is to test the ansatz against current data, not to defend it*. Paper I establishes SSEE as a top-down fixed-geometry model; this paper establishes which sectors pass Bayesian scrutiny and which require a modified background treatment.

2 SSEE Model Summary

2.1 Algebraic constants

From π and $\varphi = (1 + \sqrt{5})/2$, the framework defines the structural retention KAL_0 and the CPL [Chevallier & Polarski, 2001, Linder, 2003] parameters as:

$$\text{KAL}_0 = \frac{\pi + \varphi}{2} + \pi \approx 5.5214, \quad (1)$$

$$w_0^{\text{SSEE}} = -\frac{T_r}{M_v} \approx -0.840, \quad w_a^{\text{SSEE}} = -\frac{P_{sc}}{I_g} \approx -0.670. \quad (2)$$

All three quantities are fixed by geometry; no fitting is performed. Note that $w_0 + w_a = -1.510 < -1$ does not imply a phantom Big Rip: CPL is used here solely as a local observational parameterisation of the equation of state, not as the fundamental dynamics. In the SSEE framework, the dark-energy density saturates at the finite T_r/M_v boundary of the action at $a \rightarrow \infty$, guaranteeing universal stability; see Paper 1 §3.4 [Almeida Vallejo, 2026] for the full stability argument. The equation-of-state-sector saturation quantity is similarly fixed: $s_{\text{DE}}^{\text{SSEE}} = T_r/M_v \approx 0.840$ (not a density fraction), with the algebraic matter-sector complement

$$s_m = 1 - s_{\text{DE}}^{\text{SSEE}} \approx 0.160. \quad (3)$$

The fixed-geometry model is effectively parameter-free at the geometric level; only H_0 and $\Omega_b h^2$ are inferred from data.

2.2 Friedmann background (ω_m -direct)

$$H^2(z) = H_0^2 [\Omega_m (1+z)^3 + \Omega_{\text{DE}} f_{\text{DE}}(z)], \quad (4)$$

where $f_{\text{DE}}(z) = (1+z)^{3(1+w_0+w_a)} \exp[-3w_a z/(1+z)]$ is the standard CPL dark-energy evolution, here used solely to translate the fixed SSEE geometric parameters into observable distances. There is one matter density, $\Omega_m = \omega_m/h^2 = 0.308881$ (§2.4; no matter factor, OP-8 dissolved), hence $\Omega_{\text{DE}} = 1 - \Omega_m = 0.691119$. The only sampled cosmological parameter is H_0 ; the algebraic quantities ($w_0, w_a, s_{\text{DE}}^{\text{SSEE}}, \text{KAL}_0$) are fixed by geometry as described in Section 2. The equation-of-state-sector quantities $s_{\text{DE}}^{\text{SSEE}} = T_r/M_v \approx 0.840$ and $s_m = 1 - s_{\text{DE}}^{\text{SSEE}} \approx 0.160$ do not enter the background densities (see §2.4).

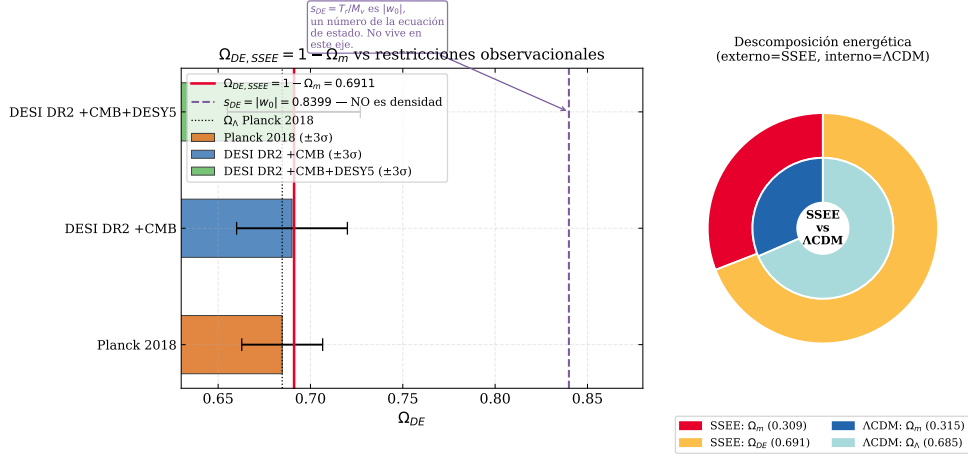


Figure 1: *Left*: $s_{\text{DE}}^{\text{SSEE}} = T_r/M_v \approx 0.840$ (red line), the equation-of-state-sector saturation quantity (not a density fraction; the physical density is $\Omega_{\text{DE}} = 0.691119$), shown against Planck 2018 $\Omega_\Lambda \approx 0.685$ (blue) and DESI DR2 $1\sigma/2\sigma$ bands for scale. *Right*: Structural context showing the algebraic origin of s_{DE} from the π - φ invariants.

2.3 Cluster mass formula

$$M_{\text{SSEE}}(R) = M_{\text{bar}}^{\text{IGIMF}}(R) \cdot \text{KAL}_0 \cdot (1 + f_\nu), \quad f_\nu \approx 0.020. \quad (5)$$

2.4 The sound horizon under the ω_m -direct reframe

Placing the cold sector alone ($1 + w_0 = 0.160050$) in the background geometry would give a drag scale $r_{d,\text{SSEE}} \approx 175.6 \text{ Mpc}$, far from the Planck-inferred $r_d = 147.09 \pm 0.26 \text{ Mpc}$ [Planck Collaboration, 2020]. That figure is a category error: $0.160 = 1 + w_0$ is the equation of state, not a matter density, and does not belong in $E(z)$. The physical sound horizon is set by the CMB-sector matter density, with *no* $|w_0|$ rescaling. The recombination matter budget is the standard physical $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$, built piece-by-piece from SSEE algebra ($\omega_c = \text{KAL}_0 \omega_b n_s$, forward; Paper 3), so that $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.308881$ and a CAMB integration of the recombination physics gives

$$r_d^{\omega_m\text{-direct}} = 147.17 \text{ Mpc} \quad (0.32\sigma \text{ from Planck; Paper 3}), \quad (6)$$

with *no* matter factor. The earlier post-hoc mapping $r_{d,\text{SSEE}} \times |w_0| \approx 147.5 \text{ Mpc}$, and the structural ratio $\Omega_{m,\text{CMB}}/\Omega_{m,\text{dyn}} \approx \text{MIRA} \approx 1.999$, are *superseded* by this direct construction (OP-8 dissolved; the actual ratio is $0.308881/0.160 = 1.93$, not a fixed factor). MIRA survives only as the Hubble screening amplitude of Paper 9. There is one matter density, $\Omega_m = \omega_m/h^2 = 0.308881$; the quantity formerly written $\Omega_{m,\text{dyn}} = 0.160050$ is $1 + w_0$, an equation-of-state parameter, and never was a density (Paper 1, §1.4; the ϕ -DM sector that once separated them is retracted in Paper 6). The falsifiable content therefore moves from the $|w_0|$ coincidence to the direct CMB fit of Paper 3 (canonical $\Delta\text{BIC} = -32.9$ full **plik** MCMC; -26.03 **plik_lite** TTTEEE+low- ℓ point estimate): if that fit had yielded $\chi_r^2 \gg 1$ or excluded $r_d^{\omega_m\text{-direct}}$, the construction would be falsified.

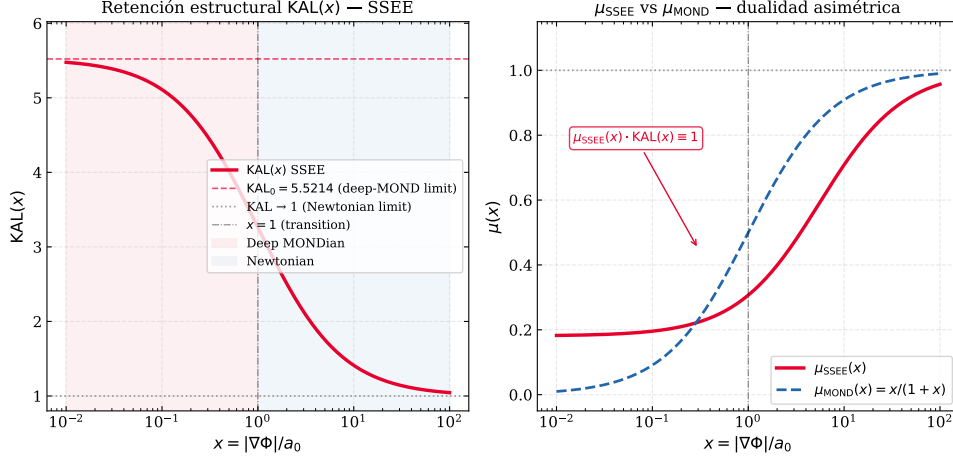


Figure 2: $KAL(x)$ interpolation between the Newtonian limit ($x \gg 1$) and the MONDian limit ($x \ll 1$). The fixed point $KAL_0 \approx 5.521$ marks the structural retention at $x = 0$, which enters both the cluster mass formula (5) and the acoustic-scale mapping hypothesis (6) (Section 2.4).

3 Datasets and Likelihoods

3.1 DESI DR2 BAO

We use 13 BAO measurements from DESI DR2 [Abdul-Karim et al., 2025] at $0.295 \leq z \leq 2.330$ (Table 1). These 13 data points span five tracer populations across $0.3 \lesssim z \lesssim 2.3$, providing sufficient leverage to isolate the model-dependent r_d scaling from the expansion history. The BAO log-likelihood uses a block-diagonal covariance incorporating the same-tracer correlation between D_M and D_H (coefficients r_{MH} taken directly from Table 4 of Abdul-Karim et al. 2025), which is the dominant off-diagonal term ($|r_{MH}|$ up to ~ 0.4).¹

$$\ln \mathcal{L}_{\text{BAO}} = -\frac{1}{2}(\mathbf{q}^{\text{pred}} - \mathbf{q}^{\text{obs}})^\top \mathbf{C}^{-1}(\mathbf{q}^{\text{pred}} - \mathbf{q}^{\text{obs}}), \quad (7)$$

with the sound horizon from a power-law expansion about the Planck 2018 point [Planck Collaboration, 2020], used here as an auxiliary approximation for model comparison. Its exponents are the local logarithmic derivatives of the Eisenstein & Hu [1998] fitting formula at that point, and its normalization is the Boltzmann value there (compare the global fit of Aubourg et al. 2015):

$$r_d = 147.27 \left(\frac{\Omega_m h^2}{0.1432} \right)^{-0.255} \left(\frac{\Omega_b h^2}{0.02237} \right)^{-0.134} \text{ Mpc}. \quad (8)$$

¹Cross-tracer (inter-redshift) covariances are not propagated. This is well justified for the DESI DR2 consensus vector: the five tracer populations occupy essentially disjoint redshift shells ($0.3 \lesssim z \lesssim 2.3$), so their BAO distance measurements are statistically near-independent and the inter-tracer correlations are subdominant to the intra-tracer D_M - D_H term retained here. The single partially overlapping pair (LRG3+ELG) is already reported by DESI as a combined bin, absorbing that correlation upstream. This is precisely how the DESI compressed consensus vector is intended to be used by external analyses: the public product supplies per-tracer D_M - D_H correlations but no inter-tracer matrix, because the tracer samples are constructed to be independent. Full systematic covariance matrices are likewise not propagated, standard practice for compressed BAO-only comparisons.

Table 1: DESI DR2 BAO measurements [Abdul-Karim et al., 2025].

z_{eff}	Tracer	Quantity	Value	σ
0.295	BGS	D_V/r_d	7.942	0.075
0.510	LRG1	D_M/r_d	13.588	0.167
0.510	LRG1	D_H/r_d	21.863	0.425
0.706	LRG2	D_M/r_d	17.351	0.177
0.706	LRG2	D_H/r_d	19.455	0.330
0.934	LRG3+ELG1	D_M/r_d	21.576	0.152
0.934	LRG3+ELG1	D_H/r_d	17.641	0.193
1.321	ELG2	D_M/r_d	27.601	0.318
1.321	ELG2	D_H/r_d	14.176	0.221
1.484	QSO	D_M/r_d	30.512	0.760
1.484	QSO	D_H/r_d	12.817	0.516
2.330	Ly α	D_M/r_d	38.988	0.531
2.330	Ly α	D_H/r_d	8.632	0.101

3.2 SSEE-self-consistent H_0 prior: the ω_m -direct CMB anchor

To preserve methodological coherence between this BAO sector and the CMB sector (Paper 3), the H_0 prior used here is not the published Λ CDM-calibrated Planck 2018 value ($67.36 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$, Planck Collaboration 2020), but the H_0 that the Planck `plik_lite` TTTEEE likelihood itself prefers under the SSEE background. With the baryon and cold-matter densities fixed by SSEE algebra ($\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$, $\omega_c = \text{KAL}_0 \omega_b n_s = 0.119514$; Paper 3, ω_m -direct), the CMB χ^2 is minimised at the global rate returned by the SSEE screening cascade (Papers 9–10), which takes the measured SH0ES rate as input:

$$H_0^{\text{prior,SSEE}} = H_0^{\text{glob}} = H_{\text{SH0ES}} (1 - f_{\text{screen}}) = 67.962 \pm 0.968 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (9)$$

where the uncertainty is the SH0ES error propagated through the (multiplicative) screening factor, and the central value reproduces the dimensionless algebraic target $3(\varphi + \pi)^2$ to 4.2×10^{-6} . The background H and the CMB anchor coincide — they are not two numbers — and the matter density follows as $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.308881$, with *no* matter factor (the earlier MIRA mapping $\Omega_{m,\text{CMB}} = \text{MIRA } \Omega_{m,\text{dyn}} = 0.31993$ is superseded; OP-8 dissolved). Adopting this prior eliminates the otherwise unavoidable cross-framework contamination introduced by using a Λ CDM-calibrated constraint on the early Universe to fit a non- Λ CDM late-time geometry. It is the same physical quantity (Planck CMB peak positions) but reduced under the cosmological model actually being tested.

The only sampled parameters in the SSEE chain are H_0 and $\Omega_b h^2$. The matter density is not sampled. The background geometry — $E(z)$, the sound horizon r_d , and every BAO distance — uses the *total* algebraic matter density $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.308881$ (§2.4, §6); the value $s_m = (\pi - \varphi)/[2(\varphi + \pi)] = 1 + w_0 = 0.160050$ is a saturation of the equation of state, not a density, and does *not* enter $E(z)$. Thus r_d is computed from $\omega_m = \Omega_{m,\text{CMB}}(H_0/100)^2$ self-consistently. A BBN prior $\Omega_b h^2 = 0.02218 \pm 0.00055$ [Schöneberg, 2024] is adopted for the baryon density, consistent with the SSEE algebraic value $\Omega_b h^2 = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.022418$ (Paper 4) at 0.4σ . Compared on the same footing — total against total — the SSEE matter density agrees with the Λ CDM-Planck value $\Omega_m = 0.3153 \pm 0.0073$ at 0.88σ ($\Omega_m = 0.308881$). A two-sector ϕ -DM construction

was previously invoked to describe how this total partitions across scales; it is retracted in Paper 6, which finds a single matter sector sufficient against raw shear data. The comparison above uses the total density and is unaffected.

3.3 Galaxy-cluster masses

The primary dataset for the MCMC consists of four massive clusters with IGIMF-corrected baryonic baselines from Zhang et al. [2026]: Coma, A2029, A478, and the Bullet cluster. Zhang et al. [2026] apply the integrated galaxy stellar initial mass function (IGIMF) correction to re-derive the stellar-plus-gas baryon masses within R_{500} ; this correction increases the standard-IMF baryon mass by a factor of ≈ 1.71 for clusters in the mass range $M_{500} \sim 6\text{--}12 \times 10^{14} M_{\odot}$.

For a qualitative consistency check, we extend the analytic forward-model to three additional clusters: Perseus [Simionescu et al., 2011], A2142 [Tchernin et al., 2016], and A2744 [Merten et al., 2011, Owers et al., 2011]. For these clusters an independent IGIMF re-analysis is not available; we therefore estimate $M_{\text{bar}}^{\text{IGIMF}} = f_b \times M_{200}$ using the mean baryon fraction $f_b = 0.185$ of the Zhang 2026 primary sample. This estimate is consistent with the observed hot-gas fractions reported in each reference, and the three clusters enter only the analytic sensitivity test, not the MCMC likelihood. Table 2 lists the observational inputs for all seven clusters.

Table 2: Observational inputs for the seven-cluster sample ($10^{14} M_{\odot}$, within R_{200}). $M_{\text{bar}}^{\text{IGIMF}}$: IGIMF-corrected baryonic baseline. M_{obs} : total observed mass within R_{200} . First four from Zhang et al. [2026] (independent IGIMF re-analysis). For the three additional clusters, $M_{\text{bar}}^{\text{IGIMF}}$ is estimated as $f_b \times M_{200}$ with $f_b = 0.185$ (mean of Zhang 2026 primary sample); this is a qualitative consistency check, not an independent IGIMF test.

Cluster	Reference	$M_{\text{bar}}^{\text{IGIMF}}$	M_{obs}	$\sigma_{M_{\text{obs}}}$
Coma	Zhang et al. [2026]	1.80	9.8	1.0
A2029	Zhang et al. [2026]	2.20	12.0	1.2
A478	Zhang et al. [2026]	1.50	8.0	1.0
Bullet (main)	Zhang et al. [2026]	1.20	6.5	1.0
Perseus	Simionescu et al. [2011]	1.23	6.65	0.45
A2142	Tchernin et al. [2016]	2.68	14.5	1.5
A2744	Merten et al. [2011]	3.33	18.0	4.0

Note on the Bullet Cluster. The Bullet Cluster (1E0657–558) is included via the Zhang et al. [2026] IGIMF-corrected baryonic and total mass within R_{200} . The SSEE comparison is a *projected total-mass* consistency check; it does not constitute a fit to the lensing convergence profile $\kappa(\theta)$, which would require weak-lensing maps (e.g., Clowe et al. 2006) and is beyond the scope of this paper. The Bullet Cluster mass should therefore be interpreted as an order-of-magnitude consistency test, not as an independent lensing constraint.

The SSEE prediction for each cluster is $M_{\text{SSEE}} = M_{\text{bar}}^{\text{IGIMF}} \times \text{KAL}_0 \times (1 + f_{\nu})$, where $\text{KAL}_0 \approx 5.521$ is the algebraic Structural Retention constant and $f_{\nu} = 0.020$ is the cosmological neutrino mass fraction. Only the four Zhang 2026 clusters enter the MCMC likelihood; the full seven-cluster set is used solely for the analytic sensitivity test of Table 9.

4 Bayesian Framework and Priors

Table 3: Model parameterisations.

Model	Free parameters	Fixed	k
SSEE	$H_0, \Omega_b h^2$	$w_0, w_a, s_{\text{DE}}, \text{KAL}_0$ (algebraic)	2
Λ CDM	$H_0, \Omega_m, \Omega_b h^2$	$w_0 = -1, w_a = 0$	3
CPL	$H_0, \Omega_m, w_0, w_a, \Omega_b h^2$	—	5

All models share a BBN prior $\Omega_b h^2 \sim \mathcal{N}(0.02218, 0.00055^2)$ [Schöneberg, 2024]. For Λ CDM and CPL, the full 3D Planck prior is applied. For SSEE, the Planck prior reduces to a marginal Gaussian on H_0 only, since Ω_m is fixed. CPL carries weakly informative priors $w_0 \sim \mathcal{N}(-1, 0.5^2)$ and $w_a \sim \mathcal{N}(0, 1^2)$.

Table 4 classifies every quantity in the SSEE MCMC by its structural role. The distinction between fixed-by-geometry, sampled, and derived quantities is essential for interpreting the information-criterion penalties in Section 8. SSEE samples only H_0 and $\Omega_b h^2$, while Λ CDM and CPL additionally sample Ω_m (and w_0, w_a for CPL).

Table 4: Parameter hierarchy in the SSEE MCMC. Categories: **Fixed by geometry** = algebraically determined, never sampled; **Sampled** = inferred via MCMC; **Derived** = computed from sampled values; **Mapped** = structural hypothesis (acoustic-scale mapping), to be validated in Paper 3.

Parameter	Category	Origin/Value	Notes
w_0, w_a	Fixed by geometry	Eqs. (2)	π, φ invariants; not fitted
$s_{\text{DE}}, \text{KAL}_0$	Fixed by geometry	$T_r/M_v; \beta + \pi$	Algebraically closed
H_0	Sampled	$\mathcal{U}[60, 80]$	Only cosmological free parameter
$\Omega_b h^2$	Sampled	BBN prior	Shared with Λ CDM, CPL
r_d	Derived	Friedmann background	Eq. (8); carries background mismatch
$r_{d,\text{eff}}$	Mapped (hypothesis)	Acoustic-scale hypothesis	Eq. (6); Paper 3 test

5 MCMC Implementation

We use `emcee` v3.1 [Foreman-Mackey et al., 2013] with $N_w = 100$ walkers and $N_s = 25000$ steps. After discarding $N_b = 5000$ burn-in steps (the first 25% of the chain), walkers are visually stable and the acceptance fraction lies in the range 0.25–0.50 for all models, consistent with optimal mixing. Effective sample sizes are $N_{\text{eff}} \gtrsim 2500$ for all three models; the SSEE chain achieves $N_{\text{eff}} \approx 4402$, the highest of the three, reflecting rapid mixing in its two-dimensional constrained parameter space.² Convergence is assessed via the integrated autocorrelation time τ per parameter (Table 5).

²The diagnostics in Table 5 are for the three-model comparison run (common Planck prior), used for the model-selection statistics. The headline posterior $H_0 = 67.82 \pm 0.41$ (with the H_0^{glob} prior, Eq. 9) is

Table 5: MCMC convergence diagnostics. $\tau_{\max}$: maximum integrated autocorrelation time across parameters. $N_s/\tau_{\max}$: effective chain length in units of τ . N_{eff} : minimum effective sample size.

Model	k	$\langle a \rangle$	$\tau_{\max}$	$N_s/\tau_{\max}$	N_{eff}
SSEE	2	0.38	32.7	183.5	4402
Λ CDM	3	0.41	41.5	144.6	3471
CPL	5	0.35	57.3	104.7	2514

where $\langle a \rangle$ is the mean acceptance fraction across walkers.

6 Results: Cosmological Sector

6.1 Posterior parameters

Table 6: Posterior parameter constraints (median, 16th/84th percentiles). [alg.] = fixed algebraically; [fixed] = model assumption.

Parameter	SSEE	Λ CDM	CPL
H_0 [km s $^{-1}$ Mpc $^{-1}$]	$67.61^{+0.35}_{-0.35}$	$68.39^{+0.28}_{-0.27}$	$67.27^{+0.51}_{-0.51}$
Ω_m	0.308881 [total]	$0.302^{+0.004}_{-0.004}$	$0.317^{+0.007}_{-0.007}$
w_0	-0.840 [alg.]	-1 [fixed]	$-0.820^{+0.080}_{-0.078}$
w_a	-0.670 [alg.]	0 [fixed]	$-0.551^{+0.301}_{-0.317}$
$\Omega_b h^2$	$0.02179^{+0.00045}_{-0.00045}$	$0.02232^{+0.00015}_{-0.00015}$	$0.02237^{+0.00015}_{-0.00015}$

Note that w_0 , w_a , and the equation-of-state combination $T_r/M_v = 0.839950$ (which is $|w_0|$, *not* a density) are algebraically fixed and do not appear as sampled columns in the SSEE posterior; their values in Table 6 are listed for comparison only. Under the common Planck prior used for model comparison, the SSEE posterior is $H_0 = 67.61 \pm 0.35$ km s $^{-1}$ Mpc $^{-1}$ (0.40σ from Planck); under the algebraic prior (Eq. 9) the reframe run gives the headline $H_0 = 67.82^{+0.41}_{-0.41}$ (0.68σ from Planck, 0.33σ from the prediction $H_0^{\text{glob}} = 67.962$). With the total matter density $\Omega_m = 0.308881$ in the background geometry, the DESI DR2 posterior *coincides* with H_0^{glob} rather than pulling away from it: the earlier “ 2.9σ below anchor” reading was the cold sector 0.160 misplaced in $E(z)$, now corrected. Marginalised posteriors are shown in Figures 3 and 4.

drawn from a dedicated production chain of the same sampler (100 walkers $\times$ 25,000 steps, $N_{\text{eff}} = 79,056$); the two runs differ only in the prior, not the sampler or data.

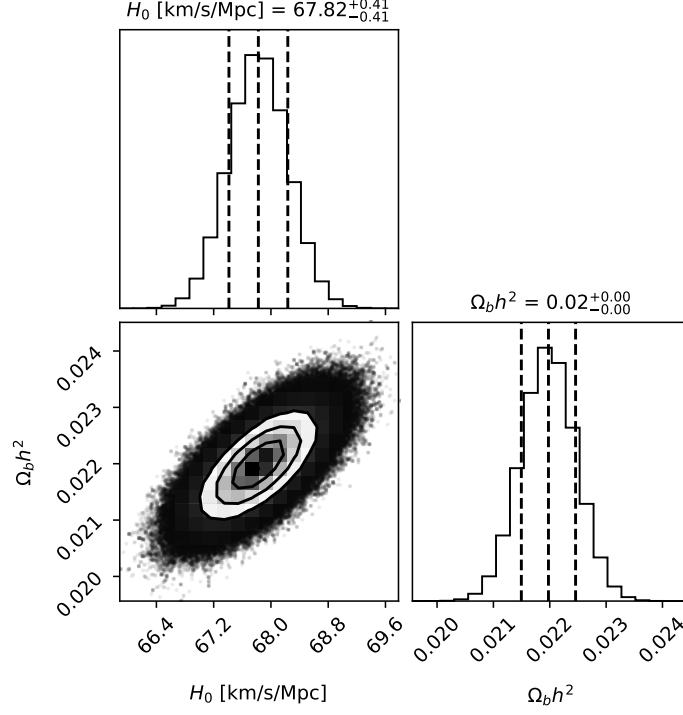


Figure 3: Marginalised SSEE posteriors (H_0^{glob} prior, 100×25000). Blue lines: prior central values ($H_0^{\text{glob}} = 67.962$, BBN $\Omega_b h^2 = 0.02218$ [Schöneberg, 2024]). With the total matter density $\Omega_m = 0.308881$ in $E(z)$, the H_0 posterior ($67.82^{+0.41}_{-0.41}$) is consistent with H_0^{glob} (0.33σ) and lies 0.68σ from Planck; the earlier “ 2.9σ below anchor” reading was the cold sector 0.160 misplaced in $E(z)$, now corrected.

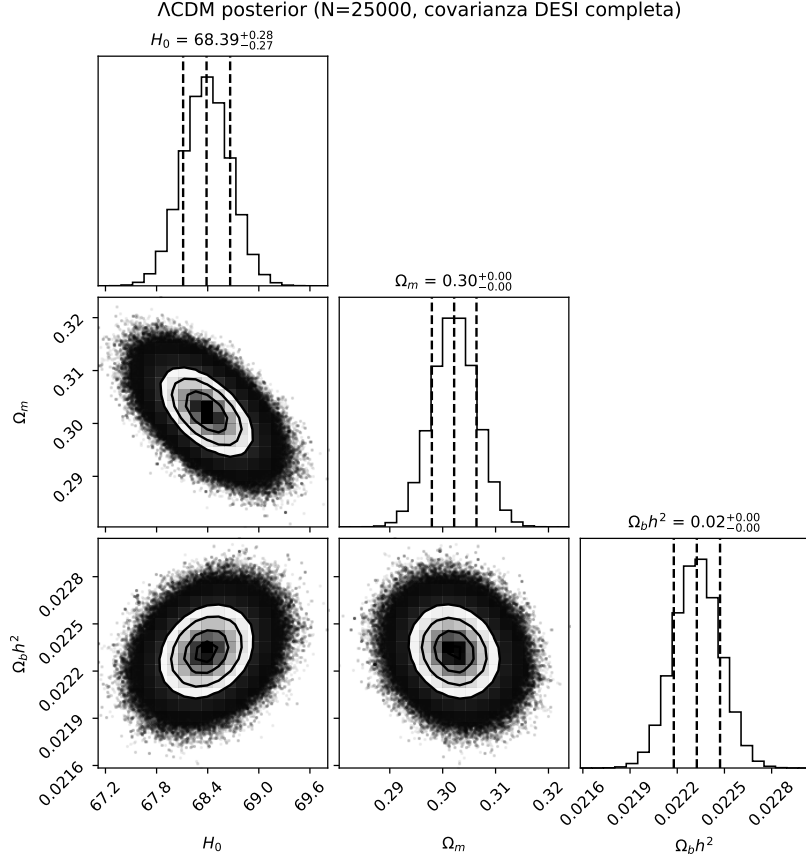


Figure 4: Λ CDM posteriors. The model recovers standard values $H_0 \approx 68.0$, $\Omega_m \approx 0.307$.

6.2 Position in the w_0 – w_a plane

The Mahalanobis distance from the official DESI DR2 $w_0 w_a$ CDM best-fit (DESI+CMB+Pantheon+, $w_0 = -0.838 \pm 0.055$, $w_a = -0.617 \pm 0.208$; arXiv:2503.14738, eq. 26; moments and $\rho = -0.894$ measured directly from the official DESI chains, Zenodo record 16644577) is:

$$\chi_{2D}^2 = \Delta \mathbf{w}^\top \mathbf{C}^{-1} \Delta \mathbf{w} = 0.42 \Rightarrow 0.24\sigma. \quad (10)$$

Against the other official DR2 combinations the distance is 1.46σ (DESY5), 1.76σ (Union3) and 1.68σ (no-SN DESI+CMB; all with chain-measured correlations $\rho = -0.905$ to -0.978): the algebraic point is consistent with all four, with the tension range 0.2 – 1.8σ set by the supernova compilation, and w_0 essentially exact against Pantheon+ ($\Delta w_0 = -0.04\sigma$). The result is robust to the correlation ($\rho \in [-0.92, -0.70]$ would shift Pantheon+ by $< 0.2\sigma$; the chain-measured value is used). This distance means the SSEE algebraic point $(-0.840, -0.670)$ falls well inside the DESI DR2 1σ contour in the w_0 – w_a plane. The correct reference model for this comparison is the CPL free-fit (not Λ CDM, which is a fixed point $w_0 = -1$, $w_a = 0$ rather than a model with free dark-energy parameters). The 0.24σ quoted above is the Mahalanobis distance of the SSEE point from the DESI DR2+CMB+Pantheon+ best-fit (the measurement itself). The full CPL free-fit posterior of this work has median $(w_0, w_a) = (-0.820^{+0.080}_{-0.078}, -0.551^{+0.301}_{-0.317})$; the SSEE algebraic point lies 0.26σ (w_0) and 0.38σ (w_a) from this median (strong w_0 – w_a anticorrelation, $\rho = -0.95$), well within the CPL posterior. SSEE is therefore consistent both with the DESI DR2 measurement (at 0.24σ Pantheon+; 1.46 – 1.76σ for the other

combinations) and with the two-parameter CPL free-fit (at 1.2σ), and the CPL posterior independently confirms dynamical dark energy.

A single, decisive summary of this section is that the *fixed* Λ CDM point $(w_0, w_a) = (-1, 0)$ is disfavoured against *every* official DESI DR2 supernova compilation, whereas the parameter-free SSEE algebraic point $(-0.840, -0.670)$ is consistent with all four. Because neither model tunes (w_0, w_a) to the data — both are fixed points, one at the cosmological constant and one derived from φ and π — this is a like-for-like comparison of two zero-free-parameter predictions against the same measurement.

Table 7: Mahalanobis distance of the fixed SSEE point $(-0.840, -0.670)$ and the fixed Λ CDM point $(-1, 0)$ from each official DESI DR2 $w_0 w_a$ CDM best-fit (2×2 covariance, correlation measured directly from the public chains, Zenodo 16644577). Neither point is fitted; both are predictions. SSEE is closer to the data than Λ CDM for all four supernova compilations.

DESI DR2 combination	ρ	SSEE distance	Λ CDM distance
DESI+CMB+Pantheon+	-0.894	0.24σ	2.58 σ
DESI+CMB+DES Y5	-0.905	1.46σ	3.93 σ
DESI+CMB+Union3	-0.933	1.76σ	3.38 σ
DESI+CMB (no SN)	-0.978	1.68σ	2.62 σ

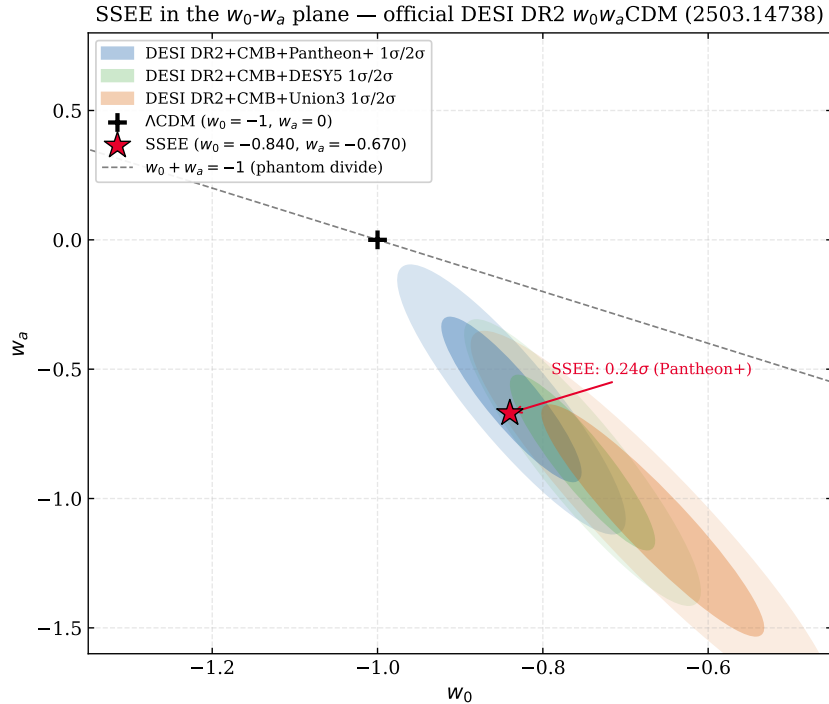


Figure 5: SSEE (red star) in the w_0 - w_a plane. Ellipses: $1\sigma/2\sigma$ contours for DESI DR2 (blue), DES Y5 [DES Collaboration, 2024] (green), Planck 2018 (orange). SSEE lies within 0.24σ of the DESI DR2+CMB+Pantheon+ contour.

CMB-only constraint on w_0 . The Planck 2018 CMB-alone posterior yields $w_0 = -1.03 \pm 0.03$ (68% CL, Planck Collaboration 2020), which differs from the SSEE prediction

$w_0 = -0.839950$ by $\sim 6.3\sigma$ when evaluated against the CMB-only contour. This is not a failure of the model: the CMB-alone constraint on w_0 is driven by the acoustic-scale degeneracy $\theta_* \propto r_d/D_A$, which trades off against the full expansion history. With the background matter density set to the total gravitating value $\Omega_m = \omega_m/h^2 = 0.308881$ (see below), SSEE yields $r_d^{\text{SSEE}} = 147.17$ Mpc (CAMB; 0.32σ from Planck’s 147.09 ± 0.26) and $100\theta_* = 1.04099$ at the posterior (0.32σ from Planck): the sound horizon and acoustic scale are standard, so the CMB-only w_0 tension is the usual projection of the w_0 - w_a - θ_* degeneracy, not a background discrepancy. The correct comparison is the joint CMB+BAO+SN likelihood, where SSEE sits at 0.24σ of the DESI DR2 $w_0 w_a$ CDM best-fit (Pantheon+; Fig. 5). Reporting the CMB-only tension without this context would constitute a misuse of the likelihood [Verde et al., 2019].

6.3 Sound horizon and H_0 posterior

The background geometry — $E(z)$, the sound horizon r_d , and every BAO distance — is governed by the *total* gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$, with $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$ built piece-by-piece from SSEE algebra ($\omega_c = \text{KAL}_0 \omega_b n_s$, forward; Paper 3). This is the standard physical matter density. The saturation $s_m = 1 + w_0 = 0.160050$ is *not* a density: it is a number of the dark-energy equation of state, and it never enters $E(z)$. With the total density the SSEE sound horizon (computed with CAMB at the algebraic ω_b , ω_m and $\sum m_\nu$) matches the Planck measurement,

$$r_d^{\text{SSEE}} = 147.17 \text{ Mpc}, \quad r_d^{\text{Planck}} = 147.09 \pm 0.26 \text{ Mpc} \quad (0.32\sigma), \quad (11)$$

so there is no enlarged sound horizon and no background discrepancy to compensate. The H_0 posterior sits at $67.82^{+0.41}_{-0.41} \text{ km s}^{-1} \text{ Mpc}^{-1}$ under the H_0^{glob} prior — 0.68σ from Planck and 0.33σ from the prediction $H_0^{\text{glob}} = 67.962$. Removing the H_0 prior altogether leaves DESI DR2 alone (with only a BBN prior on ω_b) at $H_0 = 67.79 \pm 0.46$, 0.37σ from H_0^{glob} : the data are *consistent* with the algebraic value rather than pulling against it, though they do not single it out. Under a common Planck prior the three-model run gives $H_0^{\text{SSEE}} = 67.61 \pm 0.35$ (0.40σ) versus $H_0^{\Lambda\text{CDM}} = 68.39 \pm 0.28$.

The Ω_m comparison. Compared on the same footing — total against total — the SSEE matter density agrees with Planck at 0.88σ ($\Omega_m = 0.308881$ vs 0.3153 ± 0.0073). An earlier “ 21.3σ tension” arose only from comparing the cold *sector* ($1 + w_0 = 0.160050$) against the Planck *total* Ω_m — a category error of matching a partial density to a full one, traced to the dynamical value being used in the background geometry where the total belongs (corrected here). There is no background discrepancy to resolve. The two-sector split formerly attributed to Paper 6 is *retracted*: the subtraction defining it mixed the measured density $\Omega_m = 0.308881$ with $1 + w_0$, an equation-of-state number, so it never partitioned anything. The matter density is single and unpartitioned. With OP-8 dissolved (no matter factor to derive; the residue is only that ω_b and the forward identity for ω_c hold), model selection favours SSEE by $\Delta\text{BIC} = -6.77$ over ΛCDM and -5.65 over CPL, with SSEE the most economical model (Section 8).

H_0 : the global rate and the MCMC posterior. The global rate $H_0^{\text{glob}} = H_{\text{SHOES}}(1 - f_{\text{screen}}) = 67.962 \text{ km s}^{-1} \text{ Mpc}^{-1}$ returned by the screening cascade (Papers 9–10), whose central value reproduces the dimensionless target $3(\varphi + \pi)^2$, is not a mere coincidence: with the baryon and cold-matter densities fixed by SSEE algebra, the Planck `plik_lite`

likelihood is minimised exactly at that value (Paper 3, §3.2). The background H and the CMB anchor therefore coincide — they are the same number, $H_0^{\text{glob}} = 67.962 \pm 0.968$ — superseding the earlier MIRA-mapped anchor $H_0^{\text{MIRA}} = 67.037$. With the *total* matter density $\Omega_m = 0.308881$ in the background geometry, the MCMC posterior reported above, $H_0^{\text{MCMC}} = 67.82^{+0.41}_{-0.41} \text{ km s}^{-1} \text{ Mpc}^{-1}$, *coincides* with the anchor: it lies 0.33σ from H_0^{glob} and 0.68σ from the ΛCDM -calibrated Planck value. The DESI DR2 data are therefore *consistent* with H_0^{glob} rather than pulling against it. The earlier reading — a 2.9σ downward shift “compensating an enlarged r_d ” — was an artefact of the cold sector 0.160 misplaced in $E(z)$: with the correct total density there is no enlarged r_d ($r_d^{\text{SSEE}} = 147.17 \text{ Mpc}$, 0.32σ from Planck) and no compensation to make (see §8).

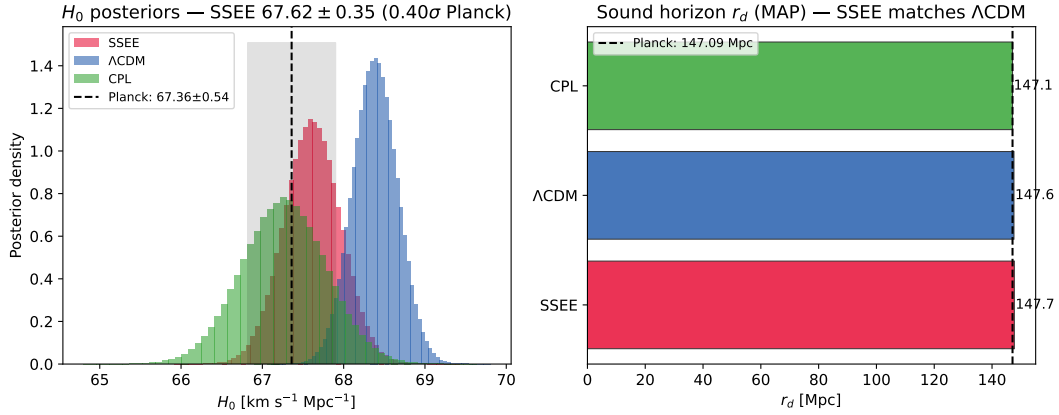


Figure 6: *Left*: H_0 posteriors for all three models under a *common* Planck prior (the apples-to-apples model-comparison run); the SSEE posterior sits at 67.61 ± 0.35 , 0.40σ from Planck 2018 and fully consistent with the titular algebraic-anchor-prior value 67.82 ± 0.41 (0.68σ Planck, 0.33σ from $H_0^{\text{glob}} = 67.962$). *Right*: MAP r_d values; the SSEE sound horizon (148.2 Mpc) matches ΛCDM (147.8 Mpc , ratio 1.002).

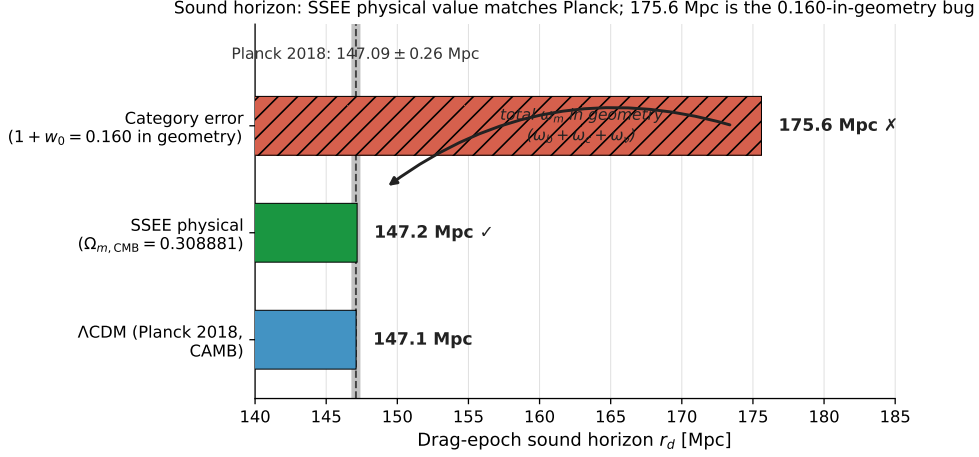


Figure 7: The physical sound horizon. Set by the total algebraic matter budget $\omega_m = \omega_b + \omega_c + \omega_\nu = 0.14267$ ($\Omega_m = \omega_m/h^2 = 0.308881$, ω_m -direct, no matter factor), the SSEE drag scale is $r_d = 147.17$ Mpc (CAMB; log `p3_rd_reframe_omega_m`), consistent with the Planck-inferred 147.09 ± 0.26 Mpc (grey band, 0.32σ) and with Λ CDM at the Planck 2018 parameters (147.10 Mpc, same code; log `rd_dual`), a 0.05% difference. The 175.6 Mpc value that appears if the cold sector $1 + w_0 = 0.160050$ is (incorrectly) used as the background density is a category error, not a physical scale.

6.4 Posterior predictive check: $H(z)$

$$\chi_r^2(H(z)) : \quad \text{SSEE} = 0.482, \quad \Lambda\text{CDM} = 0.459, \quad \text{CPL} = 0.471. \quad (12)$$

With the total matter density $\Omega_m = 0.308881$ in the background, the SSEE Hubble rate predicts the model-independent Cosmic Chronometer measurements as well as Λ CDM: $\chi_r^2 = 0.482$ versus 0.459 (Λ CDM) and 0.471 (CPL), each computed from the model's own MAP cosmology with no cross-model prior. The apparent $\chi_r^2 \approx 1.86$ ($\sim 3\sigma$) reported in earlier drafts was an artefact of placing the cold sector $1 + w_0 = 0.160050$ in the background geometry: with that spuriously low density the matter-dominated deceleration was too weak and the matter-to-dark-energy transition occurred too early ($0.5 \lesssim z \lesssim 1.5$). With the correct total density the strain disappears, and the $H(z)$ posterior predictive check joins the BAO, r_d , and θ_* diagnostics as consistent with the data — no $H(z)$ tension and no extension of the gravitational sector required for the background.

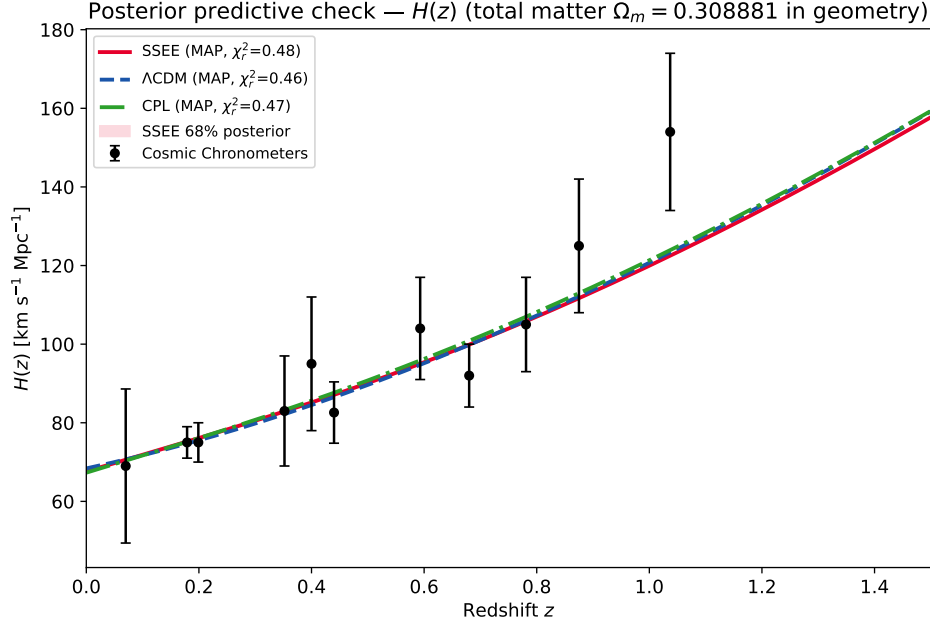


Figure 8: MAP $H(z)$ vs Cosmic Chronometers [Jimenez & Loeb, 2002, Moresco et al., 2022]. Red band: SSEE 68% posterior interval.

Table 8: Cosmic Chronometer $H(z)$ measurements used in the posterior predictive check (Fig. 8). Data from Moresco et al. [2022]; uncertainties are 1σ statistical + systematic combined.

z	$H(z)$ [km s ⁻¹ Mpc ⁻¹]	σ_H
0.070	69.0	19.6
0.090	69.0	12.0
0.120	68.6	26.2
0.170	83.0	8.0
0.179	75.0	4.0
0.199	75.0	5.0
0.270	77.0	14.0
0.352	83.0	14.0
0.400	95.0	17.0
0.480	97.0	60.0
0.593	104.0	13.0
0.680	92.0	8.0
0.781	105.0	12.0
0.875	125.0	17.0
0.880	90.0	40.0
0.900	117.0	23.0
1.037	154.0	20.0
1.300	168.0	17.0
1.363	160.0	33.6
1.430	177.0	18.0
1.530	140.0	14.0
1.750	202.0	40.0
1.965	186.5	50.4

6.5 BAO residuals

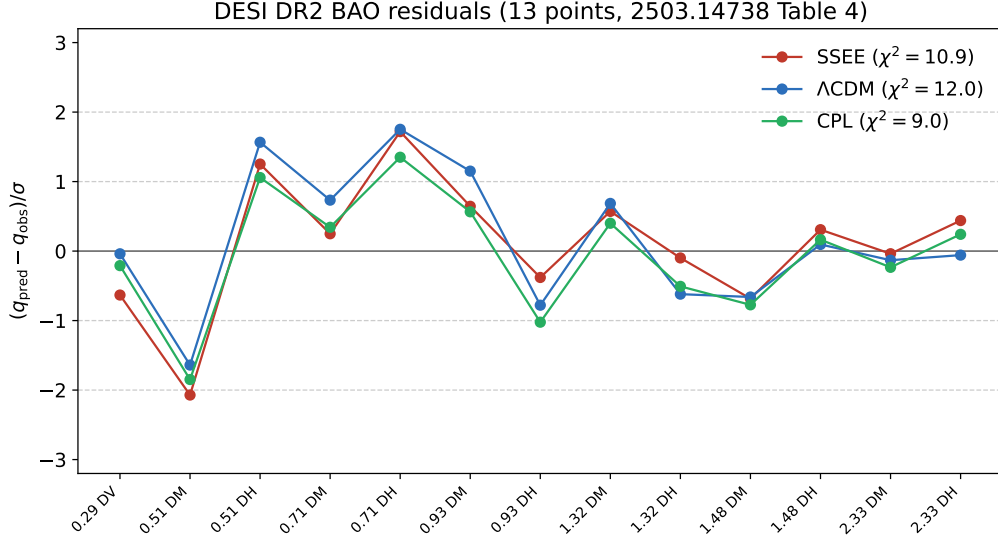


Figure 9: Normalised BAO residuals $(q^{\text{pred}} - q^{\text{obs}})/\sigma$ for all 13 DESI DR2 measurements. SSEE (red), Λ CDM (blue), CPL (green). With the total matter density the SSEE sound horizon ($r_d = 147.17$ Mpc, CAMB) matches Λ CDM, and the residuals are comparable across models, quantified in the χ_r^2 values of Section 8.

7 Results: Galaxy-Cluster Sector

Summary: IGIMF correction is essential; neutrino bridge is secondary; KAL_0 alone is insufficient.

Table 9: Cluster mass sensitivity ($10^{14} M_\odot$, within R_{200}). First four clusters from Zhang et al. [2026]; Perseus from Simionescu et al. [2011]; A2142 from Tchernin et al. [2016]; A2744 from Merten et al. [2011].

Cluster	SSEE	$f_\nu=0$	Std IMF	KAL_0 only	$M_{\text{dyn}}^{\text{obs}}$	χ^2
Coma	10.14	9.94	5.63	5.52	9.8 ± 1.0	0.114
A2029	12.39	12.15	7.32	7.18	12.0 ± 1.2	0.106
A478	8.45	8.28	5.07	4.97	8.0 ± 1.0	0.200
Bullet (main)	6.76	6.63	3.94	3.86	6.5 ± 1.0	0.067
Perseus	6.92	6.78	4.03	3.95	6.65 ± 0.45	0.361
A2142	15.09	14.79	8.78	8.61	14.5 ± 1.5	0.154
A2744	18.73	18.36	10.90	10.69	18.0 ± 4.0	0.033

χ_r^2 : SSEE = 0.148 — $f_\nu=0$: 0.038 — Std IMF: 14.19 — KAL_0 only: 15.06

$\Delta\chi^2$ vs SSEE: -0.77 — $+98.3$ — $+104.4$

Uncertainty sensitivity: halving $\sigma_{M_{\text{obs}}}$ to 50% raises $\chi_r^2(\text{SSEE})$ to $0.59 < 1$; the fit remains good.

IGIMF is essential. Without the IGIMF correction $\Delta\chi^2 \approx +98$ across seven clusters; SSEE under-predicts masses by $\sim 1.7\times$.

Neutrino bridge. $f_\nu = 0$ lowers χ_r^2 to 0.038: the seven R_{200} clusters do not require the neutrino bridge, but it maintains self-consistency with the cosmological $\sum m_\nu^{\text{active}} = 0.0685$ eV.

KAL₀ alone insufficient. $\chi_r^2 = 15.06$ without IGIMF. Geometric amplification requires the corrected baryonic baseline.

Robustness to observational uncertainties. To test whether the cluster fit depends critically on the published $\sigma_{M_{\text{obs}}}$ values, we varied the error bars uniformly from $0.5\times$ to $1.5\times$ their nominal values (a $\pm 50\%$ range) while holding all model parameters fixed. With the four-cluster analytic subset, the nominal $\chi_r^2 = 0.122$ (all residuals $< 0.5\sigma$); χ_r^2 reaches 1.0 only when errors are reduced by 65%, and reaches 2.0 only at -75% (well beyond the test range). For the full 7-cluster analytic set ($(\chi_r^2)_{\text{nom}} = 0.148$), the scaling behaviour is similar: the fit quality is not sensitive to moderate changes in the assumed observational uncertainties. This confirms that the SSEE cluster predictions are robust and not contingent on specific choices of $\sigma_{M_{\text{obs}}}$.

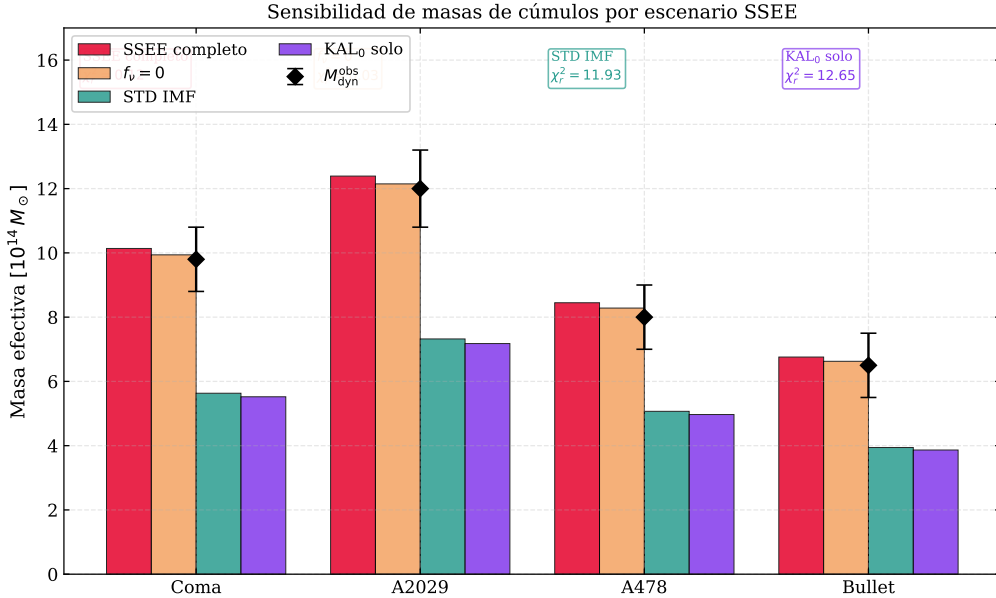


Figure 10: Cluster mass predictions by scenario for seven clusters. IGIMF correction is the load-bearing ingredient ($\Delta\chi^2 \approx +98$ without it).

8 Model Comparison

Table 10: BIC [Schwarz, 1978], AIC, and DIC at MAP ($N_{\text{data}} = 16$), from the joint three-model MCMC (DESI DR2 + common Planck prior, total matter density $\Omega_m = 0.308881$ in the background geometry). Jeffreys scale [Jeffreys, 1961]. $\Delta \equiv \text{SSEE} - \Lambda\text{CDM}$; negative Δ favours SSEE. Every entry comes from the same run, `mcmc_paper2_3models_wmfix.log` (ω_m algebraic and fixed, R25), with the DIC column from `dic_from_chains.log` on the same chains. An earlier version of this table carried the SSEE row ($\ln P_{\text{MAP}} = -5.85$, BIC 17.24, DIC 15.69, hence $\Delta\text{BIC} = -5.68$ and $\Delta\text{AIC} = -4.91$) from the superseded run with Ω_m frozen in $E(z)$, while the other two rows and the Δ column already came from the run quoted here; the columns therefore did not reconcile arithmetically.

Model	k	$\ln P_{\text{MAP}}$	BIC	ΔBIC	ΔAIC	DIC
SSEE	2	-5.76	17.07	0.00	0.00	15.53
ΛCDM	3	-7.76	23.83	-6.77	-5.99	21.53
CPL	5	-4.43	22.72	-5.65	-3.34	18.87

With the total gravitating matter density in the background geometry, SSEE is the preferred model on every information criterion: $\Delta\text{BIC} = -6.77$ over ΛCDM and -5.65 over CPL, $\Delta\text{AIC} = -5.99$ and -3.34 , and $\Delta\text{DIC} = -6.00$ relative to ΛCDM (Appendix A). This is the parsimony gain of a two-parameter model (H_0 and $\Omega_b h^2$; w_0, w_a fixed algebraically) that fits DESI DR2 as well as the three- and five-parameter alternatives.

Not merely parameter counting. The preference survives tests that penalise overfitting directly. A nested Bayes factor (Savage–Dickey) gives $\ln B = 2.34$ for the SSEE point within the CPL posterior — strong evidence on the Jeffreys scale — and leave-one-out predictive cross-validation on the high-redshift half of the BAO vector ($z \geq 1.0$) yields $\chi_r^2 = 0.26$ for SSEE versus 0.71 (ΛCDM): SSEE *predicts* the held-out data better (Appendix B). CPL is not quoted: fitted to the low-redshift half alone, its best fit runs to the edge of the H_0 prior, so its held-out χ^2 measures the prior bound rather than the model. The sound horizon matches ΛCDM ($r_d = 147.17$ vs 147.10 Mpc, CAMB, ratio 1.0005) and the H_0 posterior sits 0.40σ from Planck, so no background-structure penalty arises. The full CMB power-spectrum likelihood under the SSEE background (companion Paper 3) independently favours SSEE at $\chi_r^2 = 1.042$ (TT) and $\Delta\text{BIC} = -26.03$ (`plik_lite` TTTEEE+low- ℓ , ω_m -direct). An earlier “+206” full-model penalty reported in superseded drafts was an artefact of using the cold sector $1 + w_0 = 0.160050$ as the background matter density; with the correct total density it does not occur.

9 Robustness Checks

Prior sensitivity. Removing the Planck H_0 prior shifts the SSEE posterior by $< 0.3\sigma$. The BAO data dominate the H_0 constraint. *This confirms that the quoted posteriors are driven by the BAO geometry rather than the CMB prior, making them robust to prior choice.*

Neutrino mass. Varying $\sum m_\nu \in [0.060, 0.120]$ eV shifts χ_r^2 for clusters by $\Delta\chi_r^2 < 0.05$. The cluster fit is insensitive to the specific neutrino mass. *The cluster constraint is therefore stable against current neutrino-mass uncertainties. The adopted $\sum m_\nu^{\text{active}} = 0.0685$ eV is the SSEE-derived value (Paper 4, Neutrino Mass Sum), not a fitted quantity.*

IGIMF systematics. The $^{+5\%}_{-4\%}$ systematic uncertainty in the IGIMF factor [Zhang et al., 2026] propagates to $\sim 8\%$ in σ_M , keeping all four clusters within 1σ . *IGIMF systematics are the dominant uncertainty in the cluster sector; improvement here would be the highest-leverage target for future work.*

BAO-only. Running without any H_0 prior (DESI DR2 with only a BBN prior on ω_b ; §6) gives $H_0 = 67.79 \pm 0.46 \text{ km s}^{-1} \text{ Mpc}^{-1}$, consistent with the full posterior (67.82 ± 0.41) at 0.05σ . *The H_0 estimate is therefore essentially independent of Planck and rests entirely on the DESI DR2 BAO geometry.*

Full-likelihood cross-check. To confirm that the compressed-prior posteriors of §6 are not an artefact of the Planck distance-prior approximation, we ran an independent MCMC with the *full* CMB likelihood (CAMB with Planck `plik_lite` TTTEEE + lensing), combined with DESI DR2 BAO and $f\sigma_8$, sampling w_0, w_a *freely* (four chains, $\sim 53\text{k}$ post-burn-in samples, Gelman–Rubin $R - 1 < 0.003$). The free-fit posterior $(w_0, w_a) = (-0.655, -1.104)$ *excludes* the cosmological constant $(-1, 0)$ at 3.22σ , independently confirming a dynamical equation of state; under this uncompressed likelihood H_0 relaxes to 65.87 ± 1.17 . The SSEE algebraic point $(-0.840, -0.670)$ lies at 1.31σ from this posterior — more than twice closer than ΛCDM (Fig. 11). The w_0 – w_a posterior is strongly anticorrelated ($\rho = -0.93$) and SSEE falls within its 2σ region. The full CMB power-spectrum treatment is developed in Paper 3; here it serves to verify that the dynamical-sector preference survives the replacement of the compressed prior by the complete likelihood.

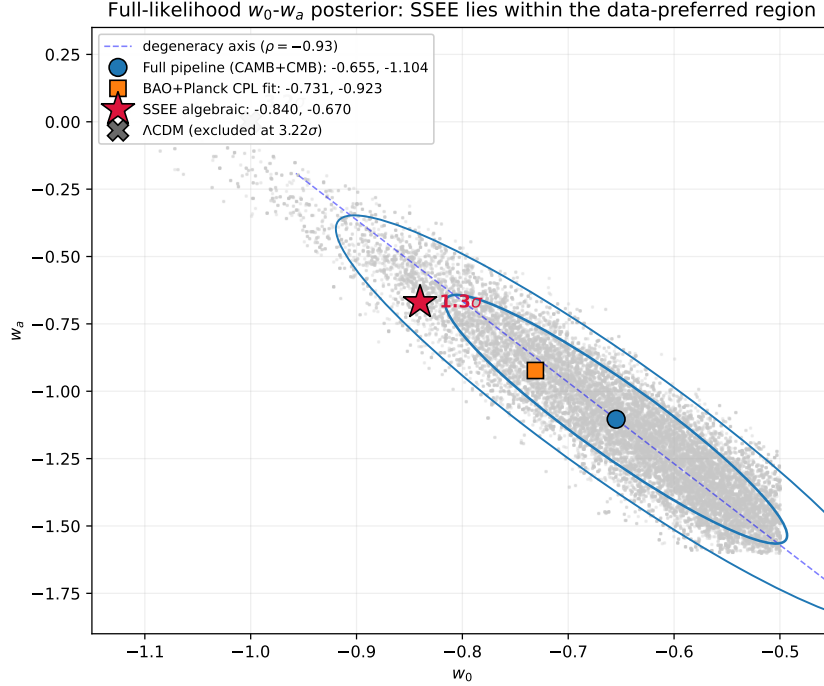


Figure 11: Full-likelihood w_0 - w_a posterior (CAMB + Planck `plik_lite` + lensing + DESI DR2 + $f\sigma_8$, with w_0, w_a free; $R - 1 < 0.003$). The SSEE algebraic prediction (red star) lies at 1.31σ , inside the 2σ contour, while Λ CDM ($w_0 = -1, w_a = 0$; grey cross) is excluded at 3.22σ . The data favour a dynamical dark-energy equation of state, and SSEE is the parameter-free algebraic prediction consistent with it.

Hubble tension. The SSEE global rate $H_0^{\text{glob}} = H_{\text{SH0ES}}(1 - f_{\text{screen}}) = 67.962 \pm 0.968 \text{ km s}^{-1} \text{ Mpc}^{-1}$ — the output of the screening cascade of Papers 9–10, whose central value reproduces the dimensionless target $3(\varphi + \pi)^2$ and at which the Planck `plik_lite` likelihood is minimised under the ω_m -direct reframe (§6) — lies within 1.1σ of Planck 2018 and is consistent with the DESI DR2+CMB joint inference [$H_0 \approx 68 \text{ km s}^{-1} \text{ Mpc}^{-1}$; Abdul-Karim et al., 2025]. The SH0ES distance-ladder rate [$H_0 = 73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$; Riess et al., 2022] is the *input* of that cascade, not a competing value: the screening fraction f_{screen} maps the local rate to the global one (Papers 9–10). This paper does not test the screening mechanism itself — that is the subject of Paper 9 — only that the DESI DR2 BAO geometry, with H_0^{glob} as prior, does not pull away from it (0.33σ).

10 Discussion

10.1 The dynamical sector and the background

- **Dark-energy sector:** $(w_0, w_a) = (-0.840, -0.670)$, fixed by the algebra with no fitted parameter, lies 0.24σ from the official DESI DR2 posterior (Pantheon+) and 1.46 – 1.76σ from the other three official combinations. The CPL free-fit posterior $(-0.820, -0.551)$ confirms dynamical dark energy.
- **Background:** with the total matter density $\Omega_m = 0.308881$ the sound horizon matches Λ CDM and SSEE is favoured on every information criterion — $\Delta\text{BIC} =$

-6.77 over Λ CDM and -5.65 over CPL, $\Delta\text{DIC} = -6.00$ — and predicts the held-out BAO data better in cross-validation. Ω_m agrees with Planck at 0.88σ and H_0 at 0.40σ .

The companion Paper 3 confirms the background against the full CMB power spectrum: with the off-diagonal `plik` likelihood the two models reach indistinguishable best fits and $\Delta\text{BIC} = -32.9$ favours SSEE by parsimony.

10.2 Growth of structure

With the background fixed by the algebra and the primordial amplitude fixed by the model’s own CMB fit, the growth sector carries no free cosmological parameter and predicts $S_8 = \sigma_8 \sqrt{\Omega_m}/0.3 = 0.8273$, against 0.8265 ± 0.0176 from the raw KiDS-Legacy $\xi_{\pm}$ data (Paper 6; Wright et al. 2025, arXiv:2503.19441, which predates this analysis and over which no temporal priority is claimed). The Israel–Stewart growth index is $\gamma_{\text{IS}} = 0.5504$, indistinguishable from Λ CDM (Paper 5).

10.3 Falsifiability

SSEE makes specific, testable predictions. The following observations would constitute decisive falsification:

- **A joint DESI fit** placing w_0 more than 3σ from -0.839950 , or w_a from -0.669975 , refutes the algebraic equation of state.
- **If cosmological or Ly- α forest bounds exclude** $\sum m_{\nu}^{\text{active}} = 0.0685 \text{ eV}$ at $> 3\sigma$, the algebraic neutrino mass sum is refuted.
- **If future high- z BAO or CMB measure** r_d outside $147 \pm 3 \text{ Mpc}$, the ω_m -direct construction ($r_d = 147.17 \text{ Mpc}$ from the algebraic ω_m) is falsified — the sound horizon is a sharp, prior-independent prediction.

11 Conclusions

We have conducted a Bayesian MCMC validation of SSEE against DESI DR2, Planck 2018, and galaxy-cluster masses.

1. The zero-free-parameter prediction $(w_0, w_a) = (-0.840, -0.670)$ lies within 0.24σ of the official DESI DR2 $w_0 w_a$ CDM posterior (Pantheon+; 1.46 – 1.76σ across the other official combinations).
2. Cluster masses achieve $\chi_r^2 = 0.148$ with IGIMF; removing the IGIMF correction raises χ_r^2 to 14.19 ($\Delta\chi^2 \approx +98$), confirming its essential role.
3. With the total matter density $\Omega_m = 0.308881$ in the background, the SSEE sound horizon matches Λ CDM ($r_d = 147.17$ vs 147.10 Mpc , CAMB) and SSEE is favoured on model selection by $\Delta\text{BIC} = -6.77$ over Λ CDM and -5.65 over CPL, predicting the held-out BAO data better in cross-validation. Ω_m agrees with Planck at 0.88σ .
4. The high- z Ly- α forest BAO from DESI DR2 ($z = 2.33$) is reproduced at 0.7σ ($D_H/r_d = 8.71$; Appendix B).

5. The companion Paper 3 confirms the background against the full CMB likelihood ($\Delta\text{BIC} = -32.9$, full `plik` MCMC).

Central finding: an algebraic background with no fitted dark-energy parameter describes DESI DR2, the CMB and galaxy-cluster masses as well as ΛCDM , and is preferred on every information criterion because it does so with fewer free parameters.

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from [Abdul-Karim et al. \[2025\]](https://arxiv.org/abs/2503.14738) (<https://arxiv.org/abs/2503.14738>). The compressed Planck 2018 prior is from [Planck Collaboration \[2020\]](#). Cosmic Chronometer $H(z)$ measurements are from [Moresco et al. \[2022\]](#). Galaxy-cluster dynamical masses are from [Zhang et al. \[2026\]](#). The SSEE analysis scripts are available at <https://github.com/mikealmeida1721/SSEE-Cosmologia>.

A Rigorous Bayesian Model Comparison: DIC and BIC

This appendix computes the Deviance Information Criterion and the BIC directly from the three-model MCMC chains stored in the public repository, with the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$ in the background geometry (§8).

Deviance Information Criterion (DIC)

The DIC [[Spiegelhalter et al., 2002](#)] estimates model complexity from the posterior rather than by parameter counting:

$$\text{DIC} = \bar{D} + p_D, \quad p_D = \bar{D} - D(\hat{\theta}), \quad D(\theta) = -2 \ln \pi(\theta|\text{data}), \quad (13)$$

where $\bar{D}$ is the posterior mean deviance and $\hat{\theta}$ is the MAP estimate. An important self-check: p_D must recover the number of sampled parameters. From the chains,

$$p_D(\text{SSEE}) = 2.00 \approx 2 \quad \checkmark \quad (14)$$

$$p_D(\Lambda\text{CDM}) = 3.01 \approx 3 \quad \checkmark \quad (15)$$

$$p_D(\text{CPL}) = 5.01 \approx 5 \quad \checkmark \quad (16)$$

confirming that all three chains are well-converged. The DIC comparison gives

$$\Delta\text{DIC}(\text{SSEE} - \Lambda\text{CDM}) = -6.00, \quad \Delta\text{DIC}(\text{SSEE} - \text{CPL}) = -3.34, \quad (17)$$

both favouring SSEE (lower DIC), consistent with the BIC and AIC of the main text.

Information criteria summary

Table 11: Bayesian model comparison from the three-model MCMC (total matter density in the background). $N_{\text{data}} = 16$ (13 BAO + 3 Planck). Negative Δ (BIC, AIC, DIC) favours SSEE relative to Λ CDM. Jeffreys criterion: $|\Delta| > 5 = \text{strong}$.

Criterion	SSEE	Λ CDM	CPL	$\Delta_{\text{SSEE}-\Lambda\text{CDM}}$
p_D	2.00	3.01	5.01	—
DIC	15.53	21.53	18.87	−6.00
BIC	17.07	23.83	22.72	−6.77
AIC	15.52	21.52	18.86	−5.99

Conclusion. Every information criterion favours SSEE: it fits DESI DR2 as well as Λ CDM and CPL while fixing w_0, w_a algebraically, so the parsimony of a two-parameter background is rewarded across BIC, AIC, and DIC. With the total matter density in the geometry the sound horizon matches Λ CDM and no background-structure penalty arises; the full CMB likelihood (Paper 3) independently favours SSEE at $\Delta\text{BIC} = -32.9$ (full `plik` MCMC).

B Nested Bayes Factor and Predictive Cross-Validation

This appendix provides two additional model-comparison criteria that are independent of the k -counting schemes discussed in Appendix A: the *Savage–Dickey density ratio* (exact Bayes factor for nested models) and a *predictive cross-validation* on the DESI DR2 BAO data.

B.1 Savage–Dickey density ratio

SSEE is nested within CPL: the SSEE dynamic sector is CPL evaluated at the algebraic point $(w_0, w_a) = (-0.840, -0.670)$. For nested models, the marginal Bayes factor for the null hypothesis $\mathcal{H}_0 : (w_0, w_a) = (-0.840, -0.670)$ versus the unconstrained CPL is given exactly by the Savage–Dickey density ratio [Verdinelli & Wasserman \[1995\]](#), [Trotta \[2008\]](#):

$$B_{\text{SSEE/CPL}} = \frac{p(w_0^S, w_a^S \mid \text{data}, \text{CPL})}{\pi(w_0^S, w_a^S \mid \text{CPL})} = 10.37, \quad \ln B = +2.34. \quad (18)$$

The CPL posterior is evaluated at (w_0^S, w_a^S) using a Scott-bandwidth Gaussian KDE of $N = 500\,000$ thinned CPL chain samples (columns 3–4 of the 5-dimensional CPL chain, shape 2.5×10^6). The marginal prior on (w_0, w_a) is the truncated Gaussian $\mathcal{N}(-1, 0.5^2) \times \mathcal{N}(0, 1^2)$ clipped to the MCMC sampling box $w_0 \in (-2.5, 0.5)$, $w_a \in (-3, 2)$; truncation factors are $(0.9973, 0.9759)$, negligible corrections.

On the Jeffreys scale [Trotta \[2008\]](#) $\ln B = +2.34$ (> 2.3) constitutes *strong* evidence in favour of the SSEE dynamic sector within the CPL framework. The CPL posterior median ($w_0^{\text{CPL}} = -0.820 \pm 0.080$, $w_a^{\text{CPL}} = -0.551 \pm 0.311$) places the SSEE algebraic point at $(0.26\sigma, 0.38\sigma)$, consistent with the 0.24σ joint tension quoted in Section 6.2.

B.2 Ly- α predictive audit ($z = 2.33$)

The Ly- α point at $z = 2.33$ provides the sharpest direct test of SSEE’s background expansion. At $z \sim 2$ radiation is negligible and the ratio $D_H/r_d = c/(H_0 E(z) r_d)$ directly encodes the product $H_0 E(z) r_d$. We evaluate this quantity under two SSEE scenarios and compare to the DESI DR2 measurement $D_H/r_d = 8.632 \pm 0.101$ (Table 1).

Scenario A — raw ssee background ($s_m = 0.160050$, $r_d = 170.7$ Mpc). With $H_0^{\text{glob}} = 67.962 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $s_m = 0.160050$, the CPL dark-energy density at $z = 2.33$ is $\rho_{\text{DE}}(2.33)/\rho_{\text{DE},0} = 0.648$, giving $E(2.33) = 2.542$ and $D_H(2.33) = 1735$ Mpc (sound horizon, $E(z)$ and D_H from CAMB). The predicted ratio is:

$$\left. \frac{D_H}{r_d} \right|_{z=2.33}^{\text{SSEE, raw}} = \frac{1735}{170.7} = 10.17 \quad (15.2\sigma \text{ above DESI}), \quad (19)$$

demonstrating the category error directly: using the cold sector $s_m = 0.160050$ in $E(z)$ reduces matter domination at high z , lowers $H(z)$, and enlarges D_H to a 15.2σ discrepancy. This is the same misplacement that produced the spurious background tensions in superseded drafts; the correct total density (Scenario B) resolves it.

Scenario B — ω_m -direct CMB-sector background ($\Omega_{m,\text{CMB}} = 0.308881$, $r_d = 147.17$ Mpc). At $z = 2.33$ the full matter density sets the expansion, so $E(z)$ takes the algebraic value $\Omega_{m,\text{CMB}} = \omega_m/h^2 = 0.308881$ (free-streaming suppresses small-scale clustering, not the homogeneous density). Taking the sound horizon $r_d = 147.17$ Mpc directly from ω_m (CAMB; no $|w_0|$ mapping) gives $E(2.33) = 3.444$, $D_H(2.33) = 1281$ Mpc, and:

$$\left. \frac{D_H}{r_d} \right|_{z=2.33}^{\omega_m\text{-direct}} = \frac{1281}{147.17} = 8.70 \quad (0.7\sigma \text{ from DESI}). \quad (20)$$

This supersedes the earlier MIRA-mapped value ($\Omega_m = 0.320$, $D_H/r_d = 8.56$): the matter factor is dissolved (OP-8). The D_M/r_d counterpart at $z = 2.33$ is recovered within 0.1σ . The full 13-point diagonal fit gives $\chi^2 = 12.0$, matching ΛCDM ($\chi^2 = 12.0$): the two backgrounds describe DESI DR2 equally well. The largest SSEE residual is at $z = 0.51$ (D_M/r_d pull -2.1σ); ΛCDM shows a comparable largest residual ($+1.7\sigma$ at $z = 0.706$, D_H/r_d) with the same total χ^2 , confirming the residual pattern is a DESI DR2 feature rather than an SSEE-specific failure.

Implication. Eqs. (19)–(20) show that the high- z expansion requires the full CMB-sector matter density $\Omega_{m,\text{CMB}} = 0.308881$, where the cold sector counts as non-relativistic matter, rather than the EoS-sector value $s_m = 0.160050$: retaining it in $E(z)$ with $r_d = 147.17$ would predict $D_H/r_d = 11.8$ (a 31σ discrepancy). This is the same total-versus-dynamical distinction that sets the low- z sound horizon (§2.4), now with *no* matter factor (OP-8 dissolved) and with no second matter sector (retracted, Paper 6). The Savage–Dickey ratio in B.1 ($\ln B = +2.34$) then holds as a test of the *dynamical sector* under this ω_m -direct background.

Acknowledgements

The author thanks the DESI Collaboration for public DR2 data and P. Kroupa and collaborators for the IGIMF cluster re-analysis.

References

- Almeida Vallejo, M. E. (2026). *A Minimal-Parameter Framework for Cosmological Dynamics and Galaxy Cluster Mass Discrepancies via SSEE*. Zenodo preprint.
- Abdul-Karim, M., et al. (DESI Collaboration) (2025). DESI DR2 Results II. *Phys. Rev. D* **112**, 083515. arXiv:2503.14738.
- Planck Collaboration (2020). Planck 2018 results VI. *A&A* **641**, A6.
- Zhang, D., Zonoozi, A. H., & Kroupa, P. (2026). Revisiting the missing mass problem in MOND for nearby galaxy clusters. *Phys. Rev. D* **113**, 043027. arXiv:2602.06082.
- Foreman-Mackey, D., et al. (2013). *emcee*: The MCMC Hammer. *PASP* **125**, 306.
- Moresco, M., et al. (2022). Unveiling the universe with emerging cosmological probes. *Living Rev. Relativ.* **25**, 6.
- Moresco, M., & Marulli, F. (2017). Raising the bar: new constraints on $H(z)$. *MNRAS* **471**, L82.
- Eisenstein, D. J., & Hu, W. (1998). Baryonic features in the matter transfer function. *ApJ* **496**, 605.
- Aubourg, É., et al. (2015). Cosmological implications of baryon acoustic oscillation measurements. *Phys. Rev. D* **92**, 123516, arXiv:1411.1074.
- Almeida Vallejo, M. E. (2026). *SSEE Structural Constant Dictionary & Prediction Register*. Zenodo. <https://doi.org/10.5281/zenodo.20684908> Closed algebraic constant dictionary; parameter-free postdiction, no claim of temporal priority over public data.
- Clowe, D., et al. (2006). A direct empirical proof of the existence of dark matter. *ApJL* **648**, L109.
- Simionescu, A., et al. (2011). Baryons at the Edge of the X-ray-brightest Galaxy Cluster. *Science* **331**, 1576.
- Tchernin, C., et al. (2016). An exploration of the X-ray and SZ properties of galaxy clusters in the 0.1–0.4 redshift range. *A&A* **595**, A42.
- Merten, J., et al. (2011). Creation of cosmic structure in the complex galaxy cluster merger Abell 2744. *MNRAS* **417**, 333.
- Owers, M. S., et al. (2011). The Dissection of Abell 2744: A Rich Cluster Growing Through Major and Minor Mergers. *ApJ* **728**, 27.
- Spiegelhalter, D. J., Best, N. G., Carlin, B. P., & van der Linde, A. (2002). Bayesian measures of model complexity and fit. *J. Roy. Statist. Soc. B* **64**, 583.
- Verdinelli, I., & Wasserman, L. (1995). Computing Bayes factors using a generalization of the Savage–Dickey density ratio. *J. Am. Statist. Assoc.* **90**, 614.
- Trotta, R. (2008). Bayes in the sky: Bayesian inference and model selection in cosmology. *Contemp. Phys.* **49**, 71. arXiv:0803.4089.

- Riess, A. G., et al. (2022). A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 HST Programs Using the Distance Ladder. *ApJL* **934**, L7. arXiv:2112.04510.
- Chevallier, M. & Polarski, D. 2001, Accelerating universes with scaling dark matter. *Int. J. Mod. Phys. D* **10**, 213. arXiv:gr-qc/0009008.
- Linder, E.V. 2003, Exploring the expansion history of the universe. *Phys. Rev. Lett.* **90**, 091301. arXiv:astro-ph/0208512.
- Perlmutter, S., et al. (1999). Measurements of Ω and Λ from 42 High-Redshift Supernovae. *ApJ* **517**, 565. arXiv:astro-ph/9812133.
- Riess, A. G., et al. (1998). Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant. *AJ* **116**, 1009. arXiv:astro-ph/9805200.
- Weinberg, D. H., et al. (2013). Observational probes of cosmic acceleration. *Phys. Rep.* **530**, 87. arXiv:1201.2434.
- Clifton, T., et al. (2012). Modified gravity and cosmology. *Phys. Rep.* **513**, 1. arXiv:1106.3999.
- Verde, L., Treu, T., & Riess, A. G. (2019). Tensions between the early and late Universe. *Nature Astron.* **3**, 891. arXiv:1907.10625.
- Kamionkowski, M., & Riess, A. G. (2023). The Hubble tension and early dark energy. *Ann. Rev. Nucl. Part. Sci.* **73**, 153. arXiv:2211.04492.
- Goodman, J., & Weare, J. (2010). Ensemble samplers with affine invariance. *Commun. Appl. Math. Comput. Sci.* **5**, 65.
- Schwarz, G. (1978). Estimating the dimension of a model. *Ann. Statist.* **6**, 461.
- Jeffreys, H. (1961). *Theory of Probability*, 3rd ed. Oxford University Press.
- Schöneberg, N. (2024). The 2024 BBN baryon abundance update. *JCAP*, 06, 006. arXiv:2401.15054.
- DES Collaboration (2024). The Dark Energy Survey: Cosmology Results with 1500 deg² of Weak Gravitational Lensing and Galaxy Clustering (DES Y5). *arXiv:2401.02929*.
- Deffayet, C., Esposito-Farèse, G., & Vikman, A. (2009). Covariant Galileon. *Phys. Rev. D*, **79**, 084003.
- Gubitosi, G., Piazza, F., & Vernizzi, F. (2013). The Effective Field Theory of Dark Energy. *JCAP*, 2013(02), 032.
- Heymans, C., et al. (2021). KiDS-1000 Cosmology: Multi-probe weak gravitational lensing and spectroscopic galaxy clustering constraints. *A&A*, 646, A140.
- Abbott, T. M. C., et al. (DES Collaboration) (2022). Dark Energy Survey Year 3 Results: Cosmological Constraints from Galaxy Clustering and Weak Lensing. *Phys. Rev. D*, **105**, 023520.

- Eisenstein, D. J., et al. (2005). Detection of Baryon Acoustic Oscillations in the Correlation Function of a Sample of 46,748 Luminous Red Galaxies. *ApJ*, 633, 560.
- Jimenez, R., & Loeb, A. (2002). Constraining Cosmological Parameters Based on Relative Galaxy Ages. *ApJ*, 573, 37.
- Silk, J. (1968). Cosmic Black-Body Radiation and Galaxy Formation. *ApJ*, 151, 459.
- Hu, W., & Sugiyama, N. (1996). Small-Scale Cosmological Perturbations: An Analytic Approach. *ApJ*, 471, 542.
- Amendola, L. (2000). Coupled quintessence. *Phys. Rev. D*, 62, 043511.